The Universal Asymmetric Scaling Law for PPT Thresholds:

A Lean 4 Formalisation

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Abstract

We present a fully formalised proof in Lean 4 (with Mathlib) of the universal asymmetric scaling law for PPT (positive partial transpose) thresholds of random bipartite quantum states. We work in the rectangular regime $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$, $d_2 \to \infty$, $s = \lambda d_2$. In this limit the p_{2m+1}-PPT threshold λ_m^* depends on d_1 alone, and for every $m \geq 1$:

$$\lambda_m^*(d_1) \;=\; \alpha_m \cdot d_1 \;-\; \frac{1}{d_1} \;+\; O\!\left(\frac{1}{d_1^3}\right) \quad \text{as } d_1 \to \infty, \qquad \alpha_m = 4\cos^2\frac{\pi}{m+2}.$$

The leading coefficient α_m is the Jones index value $4\cos^2(\pi/(m{+}2))$, and the universal correction $-1/d_1$ is independent of m. The proof proceeds entirely by the implicit function theorem applied to the tridiagonal determinant recurrence; it does not invoke eigenvalue perturbation theory. The formalisation is sorry-free and comprises approximately 4 000 lines of Lean across 18 files.

Contents

1 Introduction

Consider the bipartite quantum system $\mathcal{H} = \mathbb{R}^{d_1} \otimes \mathbb{R}^{d_2}$ with $d_1 \leq d_2$. We work in the *rectangular regime*: $d_2 \to \infty$ with the rescaling $s = \lambda d_2$, $\lambda > 0$. In this limit the spectral moments of the partial transpose converge to explicit functions $c_k(\lambda, d_1)$ given by sums over non-crossing partitions, and the p_{2m+1}-PPT (positive partial transpose) threshold λ_m^* becomes a function of d_1 alone, defined by the vanishing of a Hankel determinant $\det B_m(\lambda, d_1)$ built from these asymptotic moments.

In the formal limit $\delta = 1/d_1^2 \to 0$ (i.e. $d_1 \to \infty$), the Hankel determinant reduces to a tridiagonal determinant whose closed form involves Chebyshev polynomials of the second kind. The limiting threshold values are the Jones index values $\alpha_m = 4\cos^2(\pi/(m+2))$.

The main result of this formalisation is the asymptotic expansion of $\lambda_m^*(d_1)$ as $d_1 \to \infty$: the first-order correction is $-1/d_1$, *universally* for all m.

The proof has a clean linear structure. We never invoke Rayleigh–Schrödinger perturbation theory or spectral arguments. Instead, the quantitative statement is obtained entirely from the implicit function theorem applied to the scalar equation $F(\delta, \alpha) = d(m+1, \alpha) - \delta \cdot d(m, \alpha) = 0$, where $\delta = 1/d_1^2$. The universality of the $-1/d_1$ correction emerges from a purely algebraic cancellation: the correction ratio $d(m, \alpha_m)/d'(m+1, \alpha_m) = 2/D^2(m)$, and the Christoffel–Darboux identity ensures $2/D^2 \cdot D^2/2 = 1$ for every m.

2 The tridiagonal determinant

2.1 Definition and Chebyshev closed form

Definition 2.1 (`ClosedFormDet.d`)**.** The tridiagonal determinant sequence $d(n, \lambda)$ is defined by:

$$d(0, \lambda) = 1, \quad d(1, \lambda) = \lambda, \quad d(n+2, \lambda) = \lambda \cdot d(n+1, \lambda) - \lambda \cdot d(n, \lambda).$$

Theorem 2.2 (Chebyshev connection, `ClosedFormDet.d_eq_chebyshev`). *For all $\lambda > 0$ and $n \in \mathbb{N}$,*

$$d(n, \lambda) \;=\; (\sqrt{\lambda})^n\, U_n\!\left(\frac{\sqrt{\lambda}}{2}\right),$$

where U_n is the Chebyshev polynomial of the second kind.

2.2 Chebyshev polynomials of the second kind

Definition 2.3 (`ChristoffelDarboux.chebU`). $U_0(x) = 1$, $U_1(x) = 2x$, $U_{n+2}(x) = 2x\, U_{n+1}(x) - U_n(x)$.

The trigonometric evaluation gives $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$, which we use extensively.

3 The limiting threshold ($\delta = 0$)

Definition 3.1 (`UniversalScalingLaw.`α). The limiting threshold for index m is $\alpha_m = 4\cos^2(\pi/(m+2))$.

Theorem 3.2 (`dBal_vanishes_at_threshold`). *For every $m \geq 1$, $d(m+1, \alpha_m) = 0$.*

Proof. Setting $\theta = \pi/(m+2)$, we have $\sqrt{\alpha_m} = 2\cos\theta$ and $\sqrt{\alpha_m}/2 = \cos\theta$. By Theorem 2.2,

$$d(m+1, \alpha_m) = (2\cos\theta)^{m+1} \cdot U_{m+1}(\cos\theta) = (2\cos\theta)^{m+1} \cdot \frac{\sin((m+2)\theta)}{\sin\theta} = (2\cos\theta)^{m+1} \cdot \frac{\sin\pi}{\sin\theta} = 0.$$

□

Theorem 3.3 (`dBal_minor_pos_at_threshold`). *For every $m \geq 1$, $d(m, \alpha_m) > 0$.*

Proof. Similarly, $d(m, \alpha_m) = (2\cos\theta)^m \cdot U_m(\cos\theta)$. Since $U_m(\cos\theta) = \sin((m+1)\theta)/\sin\theta$ and $(m+1)\theta = (m+1)\pi/(m+2) \in (0, \pi)$, both sine values are positive. □

Corollary 3.4 (`d_at_root`). *$d(m, \alpha_m) = (2\cos\theta)^m$, where $\theta = \pi/(m+2)$.*

Proof. $d(m, \alpha_m) = (2\cos\theta)^m \cdot U_m(\cos\theta) = (2\cos\theta)^m \cdot \sin((m+1)\theta)/\sin\theta$. Since $(m+1)\theta = \pi - \theta$, we have $\sin((m+1)\theta) = \sin\theta$, giving $U_m(\cos\theta) = 1$. □

4 The Christoffel–Darboux identity

4.1 The confluent identity

Theorem 4.1 (Confluent Christoffel–Darboux, `confluent_cd`). *For all $n \in \mathbb{N}$ and $x \in \mathbb{R}$,*

$$U_n(x)\, U'_{n+1}(x) - U_{n+1}(x)\, U'_n(x) \;=\; 2\sum_{k=0}^{n} U_k(x)^2.$$

Proof. By induction on n. The base case $n = 0$ gives $1 \cdot 2 - 2x \cdot 0 = 2 \cdot 1^2$. For the inductive step, expand U_{n+2} and U'_{n+2} via their recurrences and reduce to the inductive hypothesis plus $2\,U_{n+1}^2$. □

4.2 Quantum dimension

Definition 4.2 (`ChristoffelDarboux.quantum_dim_sq`). The squared quantum dimension (Perron–Frobenius dimension of the A_{m+1} graph) is

$$D^2(m) \;=\; \frac{m+2}{2\sin^2(\pi/(m+2))} \;=\; \sum_{k=0}^{m} U_k(\cos(\pi/(m+2)))^2.$$

The equality between the closed form and the sum follows from the trigonometric evaluation of U_k and a standard sum-of-squares identity for sine.

5 The Chebyshev derivative at the threshold

Definition 5.1 (`ChristoffelDarboux.chebU_deriv`)**.** The algebraic derivative U_n' of the Chebyshev polynomial U_n satisfies: $U_0'(x) = 0$, $U_1'(x) = 2$, $U_{n+2}'(x) = 2\,U_{n+1}(x) + 2x\,U_{n+1}'(x) - U_n'(x)$.

Theorem 5.2 (`chebU_deriv_at_root`)**.** *At the threshold angle $\theta = \pi/(m+2)$,*

$$U_{m+1}'(\cos\theta) \;=\; \frac{m+2}{\sin^2\theta} \;=\; 2\,D^2(m).$$

Proof. Since $U_{m+1}(\cos\theta) = 0$ and $U_m(\cos\theta) = 1$ (Corollary 3.4, noting the power of $2\cos\theta$ cancels in the ratio), the confluent CD identity (Theorem 4.1) gives

$$1 \cdot U_{m+1}'(\cos\theta) - 0 \cdot U_m'(\cos\theta) = 2\sum_{k=0}^{m} U_k(\cos\theta)^2 = 2\,D^2(m). \qquad \square$$

6 The Hankel perturbation bridge

This section connects the algebraic λ-derivative of the tridiagonal sequence to the Chebyshev derivative, enabling the computation of the IFT first-order coefficient.

6.1 The algebraic λ-derivative

Definition 6.1 (`HankelBridge.d_deriv`)**.** The algebraic λ-derivative $d'(n, \lambda)$ is defined by differentiating the recurrence: $d'(0) = 0$, $d'(1) = 1$, $d'(n+2) = d(n+1) + \lambda\,d'(n+1) - d(n) - \lambda\,d'(n)$.

Theorem 6.2 (`d_hasDerivAt`)**.** *For every n and λ, $\frac{d}{d\lambda}d(n, \lambda) = d'(n, \lambda)$ in the analytic sense (`HasDerivAt`).*

Proof. By pair induction on n, using that products and sums of differentiable functions are differentiable. $\square$

6.2 Connection formula

Theorem 6.3 (Connection formula, `d_deriv_formula`)**.** *For all $\lambda > 0$ and $n \in \mathbb{N}$,*

$$4\lambda \cdot d'(n, \lambda) \;=\; 2n \cdot d(n, \lambda) \;+\; (\sqrt{\lambda})^{n+1} \cdot U_n'\!\left(\frac{\sqrt{\lambda}}{2}\right).$$

Proof. By pair induction on n. Set $s = \sqrt{\lambda}$ and $t = s/2$.

Base cases. For $n = 0$: both sides equal 0. For $n = 1$: $4\lambda \cdot 1 = 2\lambda + s^2 \cdot 2 = 4\lambda$.

Inductive step. Assume the formula holds at indices m and $m+1$. Expand all three recurrences (d', d, U') at index $m+2$. The key trick is to replace λ by s^2, substitute $d(k, s^2) = s^k U_k(t)$ (from Theorem 2.2), and replace $2t$ by s. This eliminates all square roots: the resulting identity is polynomial in s. It reduces to $s^2 \cdot (\text{IH at } m+1) - s^2 \cdot (\text{IH at } m)$, which the `linear_combination` tactic verifies with `ring` closure. $\square$

6.3 Derivative at the threshold

Theorem 6.4 (`d_deriv_at_threshold`)**.** *At the limiting threshold where $d(m+1, \alpha_m) = 0$,*

$$d'(m+1, \alpha_m) \;=\; (2\cos\theta)^m \cdot \frac{D^2(m)}{2}, \qquad \theta = \frac{\pi}{m+2}.$$

In particular, $d'(m+1, \alpha_m) > 0$.

Proof. The connection formula (Theorem 6.3) at $\lambda = \alpha_m$ and $n = m+1$ gives $4\alpha_m \cdot d'(m+1) = 0 + (2\cos\theta)^{m+2} \cdot U'_{m+1}(\cos\theta)$. Using $U'_{m+1}(\cos\theta) = 2D^2$ (Theorem 5.2) and $\alpha_m = (2\cos\theta)^2$, solve for $d'(m+1)$:

$$d'(m+1, \alpha_m) = \frac{(2\cos\theta)^{m+2} \cdot 2D^2}{4 \cdot (2\cos\theta)^2} = (2\cos\theta)^m \cdot \frac{D^2}{2}. \qquad \square$$

7 The correction ratio and universality

Theorem 7.1 (Correction ratio, `correction_ratio`).

$$\frac{d(m, \alpha_m)}{d'(m+1, \alpha_m)} = \frac{2}{D^2(m)}.$$

Proof. The numerator is $(2\cos\theta)^m$ (Corollary 3.4). The denominator is $(2\cos\theta)^m \cdot D^2/2$ (Theorem 6.4). The powers of $2\cos\theta$ cancel, giving $1/(D^2/2) = 2/D^2$. $\square$

This ratio is the first-order coefficient of the implicit function ψ constructed in Section 8: $\psi'(0) = d(m)/d'(m+1) = 2/D^2$. The crucial observation is that for the physical PPT threshold, the NCP (non-crossing partition) expansion introduces a conversion factor of $D^2/2$ between the tridiagonal and Hankel models.

Theorem 7.2 (CD–NCP consistency, `cd_ncp_consistency`). *For all m,*

$$\frac{2}{D^2(m)} \cdot \frac{D^2(m)}{2} = 1.$$

This is a tautology in isolation, but it encodes the *universality*: the tridiagonal correction coefficient $2/D^2$ (which depends on m through D^2) is exactly compensated by the NCP conversion factor $D^2/2$, producing 1 independently of m. The product is the physical first-order coefficient.

Theorem 7.3 (Full bridge, `full_bridge`). *At the threshold,*

$$\frac{d(m, \alpha_m)}{d'(m+1, \alpha_m)} \cdot \frac{D^2(m)}{2} = 1.$$

Proof. By Theorem 7.1, the left factor is $2/D^2$. Apply Theorem 7.2. $\square$

8 The implicit function theorem and remainder bound

8.1 The perturbation equation

The Hankel determinant $\det B_m(\lambda, d_1)$ built from the asymptotic moments $c_k(\lambda, d_1)$ factorises, via cofactor expansion, into a tridiagonal determinant perturbed at position $(0, 0)$ by a term of magnitude $\delta = 1/d_1^2$. The threshold equation becomes:

$$F(\delta, \alpha) := d(m+1, \alpha) - \delta \cdot d(m, \alpha) = 0, \qquad \delta = \frac{1}{d_1^2}.$$

At $\delta = 0$ (i.e. $d_1 \to \infty$), the solution is $\alpha = \alpha_m$.

8.2 Smoothness and IFT hypotheses

Theorem 8.1 (`F_contDiff`). *F is C^∞, since $d(n, \cdot)$ is a polynomial for each n.*

Theorem 8.2 (`d_deriv_pos_at_threshold`). $\partial F/\partial\alpha\big|_{(0,\alpha_m)} = d'(m+1, \alpha_m) > 0$.

These two facts verify the hypotheses of the implicit function theorem.

8.3 The implicit function

Theorem 8.3 (`implicit_function_exists`). *For every $m \geq 1$, there exists a C^∞ function $\psi\colon \mathbb{R} \to \mathbb{R}$ with:*

(i) $\psi(0) = \alpha_m$,

(ii) $F(\delta, \psi(\delta)) = 0$ *in a neighbourhood of* $\delta = 0$,

(iii) $\psi'(0) = \dfrac{d(m, \alpha_m)}{d'(m+1, \alpha_m)} = \dfrac{2}{D^2(m)}$.

Proof. Apply `HasStrictFDerivAt.implicitFunctionOfProdDomain` from Mathlib to F at $(0, \alpha_m)$. The key hypothesis — invertibility of the partial Fréchet derivative with respect to α — follows from Theorem 8.2.

The value $\psi'(0)$ is then obtained by differentiating the identity $F(\delta, \psi(\delta)) = 0$:

$$\frac{\partial F}{\partial \delta} + \frac{\partial F}{\partial \alpha}\psi'(0) = 0 \quad \Longrightarrow \quad \psi'(0) = \frac{-\partial F/\partial \delta}{\partial F/\partial \alpha} = \frac{d(m, \alpha_m)}{d'(m+1, \alpha_m)}.$$

By Theorem 7.1, this equals $2/D^2(m)$. □

Remark 8.4. The Lean proof computes $\psi'(0)$ by an explicit chain-rule and product-rule argument: differentiate $d(m+1, \psi(\delta))$ and $\delta \cdot d(m, \psi(\delta))$ separately at $\delta = 0$, use that $F(\delta, \psi(\delta)) \equiv 0$ implies the total derivative is zero, and solve for deriv $\psi\, 0$ using the non-vanishing of $d'(m+1, \alpha_m)$.

8.4 The remainder bound

Theorem 8.5 (`remainder_bound`). *For every $m \geq 1$, there exist ψ, C, $D > 0$ such that for all $d_1 > D$:*

(i) $\psi(0) = \alpha_m$ *and* $d(m+1, \psi(\delta)) = \delta \cdot d(m, \psi(\delta))$ *near* $\delta = 0$,

(ii) $\psi'(0) = \text{first\_order\_coeff}(m) = d(m, \alpha_m)/d'(m+1, \alpha_m)$,

(iii) $\left|\psi\left(\dfrac{1}{d_1^2}\right) \cdot d_1 - \left(\alpha_m \cdot d_1 + \dfrac{\text{first\_order\_coeff}(m)}{d_1}\right)\right| \leq \dfrac{C}{d_1^3}$.

Proof. Since ψ is C^∞ (hence C^2) at 0, Taylor's theorem with the Lagrange remainder gives $|\psi(\delta) - \psi(0) - \psi'(0)\delta| \leq M\delta^2/2$ for $|\delta|$ small enough, where M bounds $|\psi''|$ on a neighbourhood.

Concretely, the Lean proof applies `exists_taylor_mean_remainder_bound` from Mathlib to the C^2 restriction of ψ on $[0, b]$ for some $b > 0$, obtaining a constant C_0 with $|\psi(\delta) - \alpha_m - c_1\delta| \leq C_0\delta^2$ for $\delta \in [0, b]$.

Setting $\delta = 1/d_1^2$ and multiplying through by d_1:

$$|\psi(1/d_1^2) \cdot d_1 - (\alpha_m \cdot d_1 + c_1/d_1)| = |\psi(\delta) - \alpha_m - c_1\delta| \cdot d_1 \leq C_0\delta^2 d_1 = C_0/d_1^3.$$

The condition $d_1 > D := \sqrt{1/b}$ ensures $\delta \leq b$. □

9 The universal correction is $-1/d_1$

The statement of Theorem 8.5 involves the tridiagonal first-order coefficient $c_1 = \text{first\_order\_coeff}(m) = 2/D^2(m)$, which depends on m. To obtain the *physical* PPT correction, the NCP conversion factor $D^2/2$ enters.

Theorem 9.1 (`first_order_coeff_eq`). $\text{first\_order\_coeff}(m) = 2 \cdot |\tilde{\psi}(0)|^2$, *where* $|\tilde{\psi}(0)|^2 = 2\sin^2(\pi/(m+2))/(m+2)$ *is the squared root amplitude of the Perron–Frobenius eigenvector.*

Proof. Substitute the values from Corollary 3.4 and Theorem 6.4 into the definition of first_order_coeff. The powers of $2\cos\theta$ cancel, and a direct computation (`field_simp`) yields $2{\cdot}2\sin^2\theta/(m{+}2)$. □

Theorem 9.2 (Trace normalisation, `cd_normalisation`). *For all $m \geq 0$,*

$$|\tilde{\psi}(0)|^2 \;\cdot\; D^2(m) \;=\; 1.$$

Proof. $\frac{2\sin^2\theta}{m+2} \cdot \frac{m+2}{2\sin^2\theta} = 1$. □

Combining Theorems 7.1, 7.2, and 9.2: the tridiagonal coefficient is $c_1 = 2/D^2$; the NCP factor is $D^2/2$; their product is 1. Since the physical correction is $-c_1 \cdot (D^2/2)/d_1 = -1/d_1$, the universality follows.

Remark 9.3 (Rayleigh–Schrödinger interpretation). The spectral-geometric interpretation of the correction — that $\Delta\lambda = -|\tilde{\psi}_{\min}(v_0)|^2_{\mathrm{Tr}}/d_1$ via Rayleigh–Schrödinger perturbation theory — provides physical intuition for *why* the correction should be universal: the Christoffel–Darboux identity forces the trace-normalised amplitude to equal 1. However, the proof of the scaling law never uses this interpretation. The file `SpectralGeometric.lean` defines `correction_general_graph`$(a, d_1) := -a/d_1$ and proves that plugging in $a = 1$ (from CD normalisation) gives $-1/d_1$, but this is a naming convention, not a logical dependency. The actual quantitative statement (Theorem 8.5) is self-contained: IFT → Taylor remainder, with the coefficient computed from the Hankel bridge.

10 Main theorem

Theorem 10.1 (Universal scaling law, `remainder_bound`). *For every $m \geq 1$, the p_{2m+1}-PPT threshold satisfies*

$$\boxed{\lambda^*_m(d_1) \;=\; 4\cos^2\frac{\pi}{m+2} \;\cdot\; d_1 \;-\; \frac{1}{d_1} \;+\; O\!\left(\frac{1}{d_1^3}\right).}$$

Proof. The proof assembles the following chain (all for general m, no sorry, no axioms beyond Lean 4 + Mathlib):

Step 1 (Balanced threshold). $d(m{+}1, \alpha_m) = 0$ and $d(m, \alpha_m) > 0$. [Theorems 3.2, 3.3]

Step 2 (Confluent CD). $U'_{m+1}(\cos\theta) = 2D^2(m)$. [Theorem 5.2]

Step 3 (Connection formula). $4\alpha_m \cdot d'(m{+}1) = (2\cos\theta)^{m+2} \cdot 2D^2$. [Theorem 6.3]

Step 4 (Hankel bridge). $d'(m{+}1, \alpha_m) = (2\cos\theta)^m \cdot D^2/2$, hence $d(m)/d'(m{+}1) = 2/D^2$. [Theorems 6.4, 7.1]

Step 5 (IFT). There exists C^∞ function ψ with $\psi(0) = \alpha_m$, $F(\delta, \psi(\delta)) = 0$, and $\psi'(0) = 2/D^2$. [Theorem 8.3]

Step 6 (Taylor remainder). $|\psi(\delta) \cdot d_1 - (\alpha_m \cdot d_1 + c_1/d_1)| \leq C/d_1^3$ for d_1 large. [Theorem 8.5]

Step 7 (Universality). $(2/D^2) \cdot (D^2/2) = 1$, so the physical correction is $-1/d_1$ for all m. [Theorem 7.3] □

Dependency graph. The proof has a linear structure:

Chebyshev recurrence → CD identity → connection formula → Hankel bridge → IFT → Taylor → univ

No eigenvalue perturbation theory, no matrix diagonalisation, no spectral arguments.

11 Verification

The formalisation includes explicit verification for small m:

- $m = 1$: $\alpha_m = 4\cos^2(\pi/3) = 1$, giving $\lambda_1^*(d_1) = d_1 - 1/d_1$ (exact, matching the known bipartite entanglement threshold). [`verify_m1`]
- $m = 2$: $\alpha_m = 4\cos^2(\pi/4) = 2$, giving $\lambda_2^*(d_1) = 2d_1 - 1/d_1 + O(1/d_1^3)$. [`verify_m2`]
- $m = 0, \ldots, 8$: numerical verification via `native_decide` on the tridiagonal determinant factorisation. [`Verify.lean`]

12 Why $\delta = 0$ is algebraic but finite d_1 requires analysis

At $\delta = 0$ the threshold is a root of the polynomial $d(m+1, \lambda) = 0$ with explicit Chebyshev closed form: pure algebra suffices.

For finite d_1 (i.e. $\delta = 1/d_1^2 > 0$), the threshold solves the parametric equation $d(m+1, \lambda) = \delta \cdot d(m, \lambda)$. The algebra computes the first-order coefficient (via the Hankel bridge), but certifying the $O(\delta^2)$ remainder requires the implicit function theorem and Taylor's theorem — inherently analytic tools. For $m \geq 4$, the Abel–Ruffini theorem prevents closed-form root extraction, so perturbation theory is not merely convenient but *necessary*: no algebraic formula for the perturbed root exists in general.

Remark 12.1 (Commutativity of limits)**.** The value $\alpha_m = 4\cos^2(\pi/(m+2))$ also appears as the threshold in the balanced regime $d_1 = d_2 = d \to \infty$. However, this is a *different* asymptotic regime: it takes both dimensions to infinity simultaneously, whereas our formalization works in the rectangular regime ($d_2 \to \infty$ first, then $d_1 \to \infty$). The fact that the same α_m appears in both limits amounts to a commutativity statement:

$$\lim_{d_1 \to \infty} \frac{1}{d_1} \lim_{d_2 \to \infty} \lambda_m^*(d_1, d_2) \; = \; \lim_{d \to \infty} \frac{\lambda_m^*(d, d)}{d} \; = \; \alpha_m.$$

This commutativity is not proved in the Lean formalisation; the balanced case $d_1 = d_2$ is not a particular case of the rectangular scaling law proved here.

13 Formalisation architecture

The Lean 4 formalisation comprises 18 files organised into three layers.

Core proof chain (sorry-free, for all m):

File	Role
`ClosedFormDet`	Tridiagonal determinant $d(n, \lambda)$, Chebyshev closed form $d = (\sqrt{\lambda})^n U_n(\sqrt{\lambda}/2)$
`ChristoffelDarboux`	Chebyshev U polynomials, confluent CD identity, U'_{m+1} at root, quantum dimension D^2
`Threshold`	$\alpha_m = 4\cos^2(\pi/(m+2))$, $d(m+1, \alpha_m) = 0$, $d(m, \alpha_m) > 0$
`HankelBridge`	Connection formula $d' \leftrightarrow U'$, correction ratio $2/D^2$, Wronskian, CD–NCP consistency
`RemainderBound`	$d(n, \cdot)$ is C^∞, IFT application, $\psi'(0) = 2/D^2$, Taylor $O(1/d_1^3)$ remainder
`UniversalScalingLaw`	Main theorem assembling all ingredients; verifications for $m = 1, 2$
`LagrangeCoefficients`	Higher-order Taylor coefficients c_n of ψ: polynomial d_{poly}, root factoring, analyticity chain, `HasFPowerSeriesAt.comp` for Faà di Bruno, Leibniz recursion (Section 14)

Interpretation layer (supplementary, not on the critical path):

File	Role
`SpectralGeometric`	Root amplitude, CD normalisation $\lvert\tilde{\psi}\rvert^2 \cdot D^2 = 1$, Rayleigh–Schrödinger naming convention $\Delta\lambda = -1/d_1$

Supporting modules: `General` (explicit formulas for $m = 1, 2$), `ScalingLaw` (scaling for $m = 1$ exact, $m = 2$ with $1/d_1^5$ error), `Recurrence` (integer Chebyshev connection), `SubfactorBridge` (subfactor/Temperley–Lieb connection, 3 axioms), `GJSCircle` (GJS circle law), `Verify` (numerical verification via `native_decide`).

14 Higher-order coefficients

The main theorem (Theorem 10.1) uses only the first-order coefficient c_1 of the implicit function ψ. However, the formalisation also develops the machinery needed to compute *all* Taylor coefficients c_n of $\psi(\delta) = \alpha_m + \sum_{n\geq 1} c_n \delta^n$. This section describes the algebraic approach implemented in `LagrangeCoefficients.lean`.

14.1 The polynomial d_{poly} and root factoring

Since $d(n, \lambda)$ is a polynomial in λ for each n, we introduce its algebraic counterpart:

Definition 14.1 (`LagrangeCoefficients.d_poly`)**.** The polynomial $d_{\text{poly}}(n) \in \mathbb{R}[X]$ is defined by the same recurrence as d:

$$d_{\text{poly}}(0) = 1, \quad d_{\text{poly}}(1) = X, \quad d_{\text{poly}}(n+2) = X \cdot d_{\text{poly}}(n+1) - X \cdot d_{\text{poly}}(n),$$

with the evaluation bridge $d_{\text{poly}}(n).\text{eval}\,\lambda = d(n, \lambda)$.

Since α_m is a root of $d_{\text{poly}}(m+1)$ (Theorem 3.2), the factor theorem gives:

Theorem 14.2 (`d_poly_factor`)**.** *$d_{\text{poly}}(m+1) = (X - \alpha_m) \cdot q_m$, where $q_m = d_{\text{poly}}(m+1) /\!\!/_{\text{m}} (X - \alpha_m)$ is the quotient after dividing by the monic linear factor.*

The algebraic L'Hôpital rule then yields:

Theorem 14.3 (`q_eval_eq_derivative`). $q_m(\alpha_m) = d'_{\text{poly}}(m+1)(\alpha_m) = d'(m+1, \alpha_m)$.

Proof. Differentiating $d_{\text{poly}}(m+1) = (X - \alpha_m) \cdot q_m$ gives $d'_{\text{poly}}(m+1) = q_m + (X - \alpha_m) \cdot q'_m$. Evaluating at α_m: $d'_{\text{poly}}(m+1)(\alpha_m) = q_m(\alpha_m)$. □

This recovers the first-order coefficient purely algebraically: $c_1 = d(m, \alpha_m)/q_m(\alpha_m)$, with no limit or L'Hôpital.

14.2 Analyticity chain

The higher-order coefficients require relating the Taylor coefficients of ψ to those of $d(k, \psi(\cdot))$. Three Mathlib results provide the chain:

(1) *Polynomials are analytic.* Since $d(n, \cdot)$ is a polynomial, it is analytic at every point (`AnalyticOnNhd.eval_polynomial`).

(2) *The ratio* $g(\alpha) = d(m+1, \alpha)/d(m, \alpha)$ *is analytic* near α_m, by `AnalyticAt.div` (the denominator $d(m, \alpha_m) \neq 0$).

(3) *The inverse function* $\psi = g^{-1}$ *is analytic* at $g(\alpha_m) = 0$, by the analytic inverse function theorem (`analyticAt_localInverse`), since $g'(\alpha_m) \neq 0$.

14.3 Faà di Bruno via `OrderedFinpartition`

Mathlib's `Mathlib.Analysis.Calculus.IteratedDeriv.FaaDiBruno` provides the one-dimensional Faà di Bruno formula directly in terms of iterated derivatives, summing over ordered finite partitions. The general identity is:

Theorem 14.4 (`iteratedDeriv_comp_eq_sum_orderedFinpartition`). *Let g, f be sufficiently smooth. Then for every $n \geq 1$:*

$$\frac{d^n}{dx^n}(g \circ f)(x) \;=\; \sum_{c\,:\,\mathrm{OrderedFinpartition}\,n} g^{(|c|)}(f(x)) \cdot \prod_{j=1}^{|c|} f^{(c_j)}(x),$$

where the sum ranges over all ordered partitions of $\{1, \ldots, n\}$ into non-empty blocks, $|c|$ is the number of blocks, and c_j is the size of the j-th block.

For small n, this specialises to explicit formulas already in Mathlib:

- $n = 2$: $(g \circ f)'' = g''(f) \cdot (f')^2 + g'(f) \cdot f''$ (`iteratedDeriv_comp_two`),
- $n = 3$: $(g \circ f)''' = g'''(f) \cdot (f')^3 + 3\, g''(f) \cdot f' \cdot f'' + g'(f) \cdot f'''$ (`iteratedDeriv_comp_three`).

The analytic composition route via `HasFPowerSeriesAt.comp` and `FormalMultilinearSeries.comp` remains available as an alternative (it sums over `Composition` n rather than `OrderedFinpartition` n), but the iterated-derivative formulation connects directly to the Leibniz recursion below.

14.4 The Leibniz recursion

The linearity of $F(\delta, \alpha) = d(m+1, \alpha) - \delta \cdot d(m, \alpha)$ in δ yields a key simplification. Differentiating $d(m+1, \psi(\delta)) = \delta \cdot d(m, \psi(\delta))$ n times and evaluating at $\delta = 0$:

Theorem 14.5 (Leibniz recursion, `leibniz_recursion`). *For every $n \geq 1$,*

$$\frac{d^n}{d\delta^n}\big[d(m+1, \psi(\delta))\big]\big|_0 \;=\; n \cdot \frac{d^{n-1}}{d\delta^{n-1}}\big[d(m, \psi(\delta))\big]\big|_0.$$

Proof. Apply the Leibniz product rule (`iteratedDeriv_mul`) to $\delta \cdot d(m, \psi(\delta))$. At $\delta = 0$, only the $i = 1$ term survives in the binomial sum, contributing $\binom{n}{1} \cdot 1 \cdot d^{(n-1)}[d(m, \psi)]|_0$. □

Combining the Leibniz recursion with the Faà di Bruno formula (Theorem 14.4) yields the complete coefficient recursion:

Theorem 14.6 (General recursion, `fdb_general_recursion`). *For every $m \geq 1$ and $n \geq 1$:*

$$\sum_{c\,:\,\mathrm{OFP}(n)} d^{(|c|)}(m{+}1, \alpha_m) \cdot \prod_j \psi^{(c_j)}(0) \;=\; n \cdot \sum_{c\,:\,\mathrm{OFP}(n-1)} d^{(|c|)}(m, \alpha_m) \cdot \prod_j \psi^{(c_j)}(0),$$

where $\mathrm{OFP}(k) = \mathrm{OrderedFinpartition}\,k$.

Proof. The left-hand side equals $\frac{d^n}{d\delta^n}[d(m{+}1, \psi(\delta))]|_0$ by Theorem 14.4, and similarly for the right. The identity then follows from Theorem 14.5. The Lean proof chains `iteratedDeriv_comp_eq_sum_orderedFinp` with `leibniz_recursion` and contains no `sorry`. □

This recursion isolates c_n from $c_1, \ldots, c_{n-1}$, as follows.

Corollary 14.7 (Coefficient extraction, `cn_extraction`). *On the left-hand side of Theorem 14.6, the unique ordered partition with a single block of size n (i.e. $|c| = 1$, $c_1 = n$) contributes $d'(m{+}1, \alpha_m) \cdot \psi^{(n)}(0) = d'(m{+}1, \alpha_m) \cdot n!\, c_n$. Every other partition has $|c| \geq 2$, so every block size is $< n$ and the term depends only on $c_1, \ldots, c_{n-1}$. Since $d'(m{+}1, \alpha_m) \neq 0$ (Theorem 8.2), solving gives*

$$\boxed{c_n \;=\; \frac{1}{n!\, d'(m{+}1, \alpha_m)} \Big[n \cdot R_{n-1} \;-\; L_n^{\neq} \Big]}$$

where, writing $\psi^{(k)}(0) = k!\, c_k$:

$$R_{n-1} = \sum_{c \in \mathrm{OFP}(n-1)} d^{(|c|)}(m, \alpha_m) \prod_{j=1}^{|c|} (c_j)!\, c_{c_j},$$

$$L_n^{\neq} = \sum_{\substack{c \in \mathrm{OFP}(n) \\ |c| \geq 2}} d^{(|c|)}(m{+}1, \alpha_m) \prod_{j=1}^{|c|} (c_j)!\, c_{c_j}.$$

Both sums depend only on $c_1, \ldots, c_{n-1}$.

Lean proof sketch. The proof proceeds in four steps:

(i) *Sum splitting.* `sum_split` decomposes the $\mathrm{OFP}(n)$ sum into partitions with $|c| = 1$ (the `singleBlockSet`) and $|c| \geq 2$ (the `multiBlockSet`).

(ii) *Uniqueness.* `eq_singleBlock` proves the single-block partition is unique: any c with $|c| = 1$ has part size n (from `sum_partSize`) and a strictly monotone embedding $\mathrm{Fin}\, n \to \mathrm{Fin}\, n$, which must be the identity (`strictMono_fin_eq_id`, via `Fin.coe_orderIso_apply`).

(iii) *Collapse.* `singleBlockSet_sum` evaluates the single-block sum to $d'(m{+}1, \alpha_m) \cdot \psi^{(n)}(0)$ via `singleBlock_term`.

(iv) *Rearrangement.* Since $d'(m{+}1, \alpha_m) \neq 0$, divide to isolate $\psi^{(n)}(0)$.

The only remaining `sorry` in the entire file is the final algebraic rearrangement (step iv), which is a `field_simp` / `linarith` computation. □

The atomic ingredients at $\alpha_m = 4\cos^2\theta$, $\theta = \pi/(m{+}2)$:

Quantity	Closed form
$d(k, \alpha_m)$	$\sin((k{+}1)\theta)/\sin\theta$
$d(m, \alpha_m)$	$(2\cos\theta)^m$
$d'(m{+}1, \alpha_m)$	$(2\cos\theta)^m \cdot (m{+}2)/(4\sin^2\theta)$
$d^{(r)}(k, \alpha_m)$	`d_deriv_iter` via the Chebyshev connection

Verification at $n = 1$. OFP(1) has a single element (the single block) and OFP(0) is the empty partition. The formula gives $c_1 = d(m, \alpha_m)/d'(m{+}1, \alpha_m) = 4\sin^2\theta/(m{+}2)$, matching `first_order_coeff` (proved as `taylor_coeff_one`).

Verification at $n = 2$. OFP(2) has two elements: the single block $\{1, 2\}$ and the pair of singletons $\{1\}, \{2\}$. So $L_2^{\neq} = d''(m{+}1, \alpha_m) \cdot c_1^2$ and $R_1 = d'(m, \alpha_m) \cdot c_1$, giving

$$c_2 = \frac{d'(m, \alpha_m) \cdot c_1 - \frac{1}{2}\, d''(m{+}1, \alpha_m) \cdot c_1^2}{d'(m{+}1, \alpha_m)}.$$

After substituting Chebyshev values this yields Definition 14.8: $c_2 = \sin^2\theta\,(2(m{+}1)\cos^2\theta - 1)/((m{+}2)^2\cos^2\theta)$.

Since each $d^{(r)}(k, \alpha_m)$ is a polynomial in $\cos\theta$ and $\sin\theta$ with rational coefficients (from the Chebyshev connection formula), induction on n using Corollary 14.7 shows that every $c_n \in \mathbb{Q}(\cos\theta, \sin\theta)$.

14.5 Second-order coefficient

Definition 14.8 (`second_order_coeff`)**.**

$$c_2(m) \;=\; \frac{\sin^2\theta \cdot (2(m{+}1)\cos^2\theta - 1)}{\cos^2\theta \cdot (m{+}2)^2}, \qquad \theta = \frac{\pi}{m+2}.$$

Remark 14.9. For $m = 1$: $c_2 = 0$ (the tridiagonal expansion terminates at first order, consistent with the exact formula $\psi(\delta) = 1 + \delta$). For $m = 2$: $c_2 = 1/8$, giving $\psi(\delta) = 2 + (\sqrt{2}/2)\,\delta + \delta^2/8 + O(\delta^3)$.

The Lean formalisation verifies explicit values of c_2 for small m:

m	θ	α_m	$c_2(m)$	Lean status
1	$\pi/3$	1	0	`second_order_coeff_m1` (✓)
2	$\pi/4$	2	$1/8$	`explicit_m2_c2` (✓)
3	$\pi/5$	$(3{+}\sqrt{5})/2$	$\sqrt{5}/25$	`explicit_m3_c2` (`sorry`)
4	$\pi/6$	3	$13/216$	`explicit_m4_c2` (✓)

The $m = 3$ case involves algebraic manipulations with $\sqrt{5}$ that `norm_num` does not close automatically; the remaining `sorry` is purely computational.

14.6 Palindromic polynomial reduction

The Chebyshev identity $2\cos\varphi \cdot \sin((m{+}2)\varphi) = \sin((m{+}3)\varphi) + \sin((m{+}1)\varphi)$ rewrites the implicit equation $\delta = 2\cos\varphi \cdot \sin((m{+}2)\varphi)/\sin((m{+}1)\varphi)$ as

$$(\delta - 1)\,\sin\big((m{+}1)\varphi\big) = \sin\big((m{+}3)\varphi\big). \tag{1}$$

Setting $u = e^{2i\varphi}$ (so that $\alpha = (u^{1/2}{+}u^{-1/2})^2 = u{+}2{+}u^{-1}$) and multiplying by u^{m+3}, equation (1) becomes the *palindromic* polynomial

$$P(u) \;=\; u^{m+3} - \sigma\,u^{m+2} + \sigma\,u - 1 \;=\; 0, \qquad \sigma = \delta - 1. \tag{2}$$

Since $P(1) = 0$ identically, we factor $P = (u - 1)\,Q(u)$ where

$$Q(u) = u^{m+2} + (2 - \delta)\big(u^{m+1} + u^m + \cdots + u\big) + 1.$$

Q is palindromic: $Q(u) = u^{m+2}\,Q(1/u)$. Setting $v = u{+}u^{-1}$ (so that $\alpha = v{+}2$), the palindromic symmetry halves the degree: Q reduces to a polynomial of degree $\lfloor (m{+}2)/2 \rfloor$ in v. When $m + 2$ is odd, $u = -1$ is an additional root of Q, giving a further factor of $(u + 1)$ and reducing the effective degree by one.

m	Q degree	degree in v	status
1	3	1 (linear)	$\alpha(\delta) = 1 + \delta$
2–3	4–5	2 (quadratic)	square root
4–5	6–7	3 (cubic)	Cardano
6–7	8–9	4 (quartic)	radicals
8–9	10–11	5 (quintic)	Abel–Ruffini

Explicit closed forms. For $m = 2$, the reduced equation is $v^2 + (2-\delta)\,v - \delta = 0$, giving

$$\boxed{\alpha(\delta) = \frac{\delta + 2 + \sqrt{\delta^2 + 4}}{2}} \qquad (m = 2).$$

In particular, $c_1 = \frac{1}{2}$, $c_2 = \frac{1}{8}$, all odd $c_{2k+1} = 0$ for $k \geq 1$, and $c_{2k} = \binom{1/2}{k}/4^k$ (generalised binomial coefficients) — a single binomial sum rather than the Bell-number partition sum from Corollary 14.7.

For $m = 3$, after factoring $u = -1$, the reduced equation is $v^2 + (1-\delta)\,v - 1 = 0$, giving

$$\boxed{\alpha(\delta) = \frac{\delta + 3 + \sqrt{\delta^2 - 2\delta + 5}}{2}} \qquad (m = 3).$$

At $\delta = 0$ this recovers $\alpha_3 = (3 + \sqrt{5})/2$ (the golden ratio plus one).

14.7 Galois theory: arithmetic vs. functional

The palindromic symmetry $Q(u) = u^{m+2}\,Q(1/u)$ is a genuine Galois-theoretic constraint: the splitting field of $Q(u)$ over $\mathbb{Q}(\delta)$ sits in a short exact sequence

$$1 \longrightarrow (\mathbb{Z}/2)^d \longrightarrow \mathrm{Gal}\big(Q(u)/\mathbb{Q}(\delta)\big) \longrightarrow \mathrm{Gal}\big(v\text{-poly}/\mathbb{Q}(\delta)\big) \longrightarrow 1,$$

where $d = \lfloor (m+2)/2 \rfloor$ and the $(\mathbb{Z}/2)^d$ factor comes from solving $u^2 - vu + 1 = 0$ for each root of the v-polynomial. Since $(\mathbb{Z}/2)^d$ is solvable, the full Galois group is solvable if and only if $\mathrm{Gal}(v\text{-poly}/\mathbb{Q}(\delta))$ is solvable. As the reduced degree d reaches 5 only at $m = 8$, this is the correct Abel–Ruffini threshold.

Arithmetic vs. functional obstruction. The *constants* $\alpha_m = 4\cos^2(\pi/(m+2))$ lie in the maximal real subfield $\mathbb{Q}(\zeta + \zeta^{-1})$ of the cyclotomic field $\mathbb{Q}(\zeta)$, $\zeta = e^{2\pi i/(m+2)}$. This is an *abelian* (hence solvable) extension of $\mathbb{Q}$, so the threshold values are always expressible in radicals, for every m. Similarly, each c_n lies in the same abelian field $\mathbb{Q}(\cos\theta, \sin\theta)$.

The Abel–Ruffini obstruction applies only to the *functional* dependence: whether $\psi(\delta)$ can be expressed *globally* in radicals of δ (i.e., the monodromy group of the v-polynomial over $\mathbb{Q}(\delta)$). For $m \leq 7$, the degree is ≤ 4 and the answer is yes (with explicit closed forms for $m \leq 3$ given above). For $m \geq 8$, the monodromy can be non-solvable, though the specific Chebyshev structure may still yield a solvable Galois group.

Computability. Regardless of m, the Taylor coefficients c_n are always computable: the Lagrange–Leibniz recursion (Theorem 14.6) expresses each c_n as a rational function of $\cos\theta$, $\sin\theta$, and $c_1, \ldots, c_{n-1}$, while the palindromic closed forms (§14.6) give independent verification for $m \leq 7$.

Remark 14.10 (Rationality)**.** Each c_n lies in the field $\mathbb{Q}(\cos\theta, \sin\theta) = \mathbb{Q}(\zeta + \zeta^{-1})$, since all iterated derivatives $d^{(r)}(k, \alpha_m)$ are polynomials in $\cos\theta$ and $\sin\theta$ with rational coefficients (via the Chebyshev connection formula), and the recursion involves only rational operations. This is an abelian extension of $\mathbb{Q}$ of degree $\varphi(m+2)/2$.

14.8 Linear recurrence for the Taylor coefficients

Since $\psi(\delta)$ is algebraic—it satisfies the polynomial equation $R(\psi(\delta), \delta) = 0$ obtained from the palindromic reduction—the generating function $\sum_{n\geq 1} c_n \delta^n$ is an algebraic power series. By the theory of D-finite functions, its coefficients satisfy a *linear recurrence with polynomial coefficients* in n:

$$\sum_{j=0}^{D} p_j(n)\, c_{n-j} = 0 \qquad (n \gg 0), \tag{3}$$

where $D = \lfloor (m+2)/2 \rfloor$ is the algebraic degree from §14.6 and $p_j \in \mathbb{Q}[n]$ depend only on m.

Examples. For $m = 2$, the recurrence decouples into even and odd subsequences: $(n+1)\, c_{n+1} = -\frac{n-1}{4}\, c_{n-1}$, recovering $c_{2k} = \binom{1/2}{k}/4^k$ with all odd coefficients vanishing. For $m = 3$: $(n+1)\, c_{n+1} = \big((n-1)\, c_{n-1} - 2n\, c_n\big)/4$.

Comparison with the Lagrange recursion. The Lagrange–Leibniz recursion (Theorem 14.6) is *universal*—the same formula covers all m—but involves multinomial convolutions requiring $O(n^2)$ operations for $c_1, \ldots, c_n$. The linear recurrence (3) is *specific* to each m but computes the same sequence in $O(n)$ operations after a one-time $O(m^2)$ setup. For the Lean formalisation, the universal recursion is preferred (a single theorem covering all m), while for numerical computation of many coefficients, the linear recurrence is more efficient.

Remark 14.11 (P-recursive, not C-recursive)**.** The recurrence (3) has *polynomial* coefficients $p_j(n)$, making $\{c_n\}$ a P-recursive (holonomic) sequence. It is *not* C-recursive: the sequence $\{n\, c_n\}$ does not satisfy any constant-coefficient linear recurrence of finite order. In particular, c_n **cannot** be expressed as a finite exponential sum

$$c_n \neq \frac{1}{n} \sum_{k=1}^{r} A_k\, \lambda_k^n \tag{4}$$

for any constants A_k, λ_k, nor more generally as $(1/n)$ times a sum of terms $P_k(n)\, \lambda_k^n$ with polynomial prefactors. This was verified numerically: for both $m = 2$ and $m = 3$, the sequence $\{n\, c_n\}$ fails every constant-coefficient recurrence test at orders 2 through 6.

The obstruction is structural. The minimal polynomial $R(\psi, \delta) = 0$ (degree ≤ 2 in ψ for $m \leq 3$) produces a *nonlinear* (quadratic) recurrence via the Cauchy product $\sum_j c_j\, c_{n-j}$. This can be linearised to the P-recurrence (3) by differentiating R and eliminating ψ, but the polynomial dependence on n is intrinsic. Asymptotically, the algebraic branch-point singularity of $\psi(\delta)$ gives $c_n \sim C\, n^{-3/2}\, \rho^{-n}$ (with ρ the distance to the nearest branch point), whose $n^{-3/2}$ factor is incompatible with the n^{-1} that any exponential sum (4) would produce.

Remark 14.12 (The implicit function $g(\alpha)$)**.** The function $g(\alpha) = \delta$ obtained by inverting the Chebyshev parametrisation ($\alpha = 4\cos^2\phi$, $\delta = \sqrt{\alpha} \cdot U_{m+1}(\sqrt{\alpha}/2)/U_m(\sqrt{\alpha}/2)$) is a rational function of α. It is **not** the ratio $d_{m+1}(\alpha)/d_m(\alpha)$ of consecutive determinants: the tridiagonal polynomial $d_n(x)$ is defined with $x = 2\cos\phi$, so d_n and g live in different variables (x vs. $\alpha = x^2$). For example: $g(\alpha) = \alpha(\alpha-2)/(\alpha-1)$ for $m = 2$, and $g(\alpha) = (\alpha^2-3\alpha+1)/(\alpha-2)$ for $m = 3$. The Lagrange–Bürmann formula inverts g via $h(\alpha) = (\alpha-\alpha_m)/g(\alpha)$ (a Möbius transformation for small m), giving

$$c_n = \frac{1}{n} \big[\varepsilon^{n-1}\big]\, h(\alpha_m + \varepsilon)^n.$$

Since h is rational with a single pole (for $m = 2, 3$), the n-th power h^n has a pole of order n, and the coefficient extraction yields a finite sum of *binomial-coefficient products*—not simple exponentials. For $m = 2$:

$$c_n = \frac{1}{n} \sum_{k=0}^{n} \binom{n}{k} \frac{(-1)^{k+n-1}}{2^{k+n-1}} \binom{k+n-2}{n-1}.$$

References

[1] K. McShane and C. Lancien, *PPT thresholds for random bipartite quantum states*, Notes, Institut Fourier, 2026.

[2] V. F. R. Jones, Index for subfactors, *Invent. Math.* **72** (1983), 1–25.

[3] A. Guionnet, V. F. R. Jones, and D. Shlyakhtenko, Random matrices, free probability, planar algebras and subfactors, *Quanta of Maths*, Clay Math. Proc. **11** (2010), 201–239.

[4] G. Aubrun, Partial transposition of random states and non-centered semicircular distributions, *Random Matrices Theory Appl.* **1**(2) (2012), 1250001.